

Performance Testing

Date	4 July 2026
Team ID	xxxxxx
Project Name	Opti Crop – Smart Agricultural Production Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Crop Prediction API (/predict)
Test Environment	Local (Flask Development Server)
Test Date	4 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single crop prediction request	1	30	Prediction generated successfully
2	Multiple prediction requests	10	60	System responds without errors
3	Continuous prediction requests	20	120	Stable performance maintained

4	Invalid input testing	5	30	Error message displayed correctly
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.85 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.10 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	Meets requirement
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	42%	Pass	Normal usage
6	Memory Utilization	< 80%	48%	Pass	Stable memory usage

Step 4: Observations & Analysis

- The application handled prediction requests efficiently with low response time.
- No request failures or server crashes were observed during testing.
- CPU and memory utilization remained within acceptable limits.
- The Flask application maintained stable performance throughout the test.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Smart Agricultural Production Optimization Engine

Welcome to OptiCrop Project

Get Started

Crop Prediction

Nitrogen (N)

Phosphorus (P)

Potassium (K)

Temperature

Humidity

pH

Rainfall

Predict Crop

Crop Prediction

Nitrogen (N)

Phosphorus (P)

Potassium (K)

Temperature

Humidity

pH

Rainfall

Recommended Crop: muskmelon